

Exploratory Data Analysis Report

Supermarket Sales Dataset

Prepared by: Priscilla Etonam Gbordjue

Internship: Data Science & Analytics, Daryl Tech & Educational Network (DTEN)

Task: Task 1, Exploratory Data Analysis Report

Dataset: Supermarket Sales Dataset (Kaggle)

Tools used: Python, Pandas, Matplotlib, Seaborn

1. Introduction

For this task, I chose a different dataset from a supermarket chain with three branches, covering 1,000 individual transactions over three months (January to March 2019). Each row records details like the branch, city, customer type, gender, product line, payment method, and the total sale amount. My aim was to explore the data for patterns around branch performance, customer behaviour, and timing, things that could realistically help a retail business plan staffing, marketing, and stock decisions.

2. Data Cleaning

Before analyzing anything, I checked the dataset's overall condition using Pandas:

1. I checked all 17 columns for missing values. There were none; every single field was fully filled in across all 1,000 rows.
2. I checked for duplicate rows using `drop_duplicates` logic. There were none either, so no rows needed to be removed.
3. I converted the 'Date' column from plain text into proper datetime format, which made it possible to extract the day of the week for later analysis.
4. I added a new 'Day of Week' column, derived directly from the Date column, to look at weekly sales patterns.

Overall, this dataset came in unusually clean, likely because it's a curated sample dataset rather than raw transaction logs, so there wasn't much cleanup work needed beyond the date conversion.

3. Key Findings

Grouping and visualizing the data surfaced four patterns I think are worth highlighting.

3.1 The Three Branches Are Fairly Evenly Matched

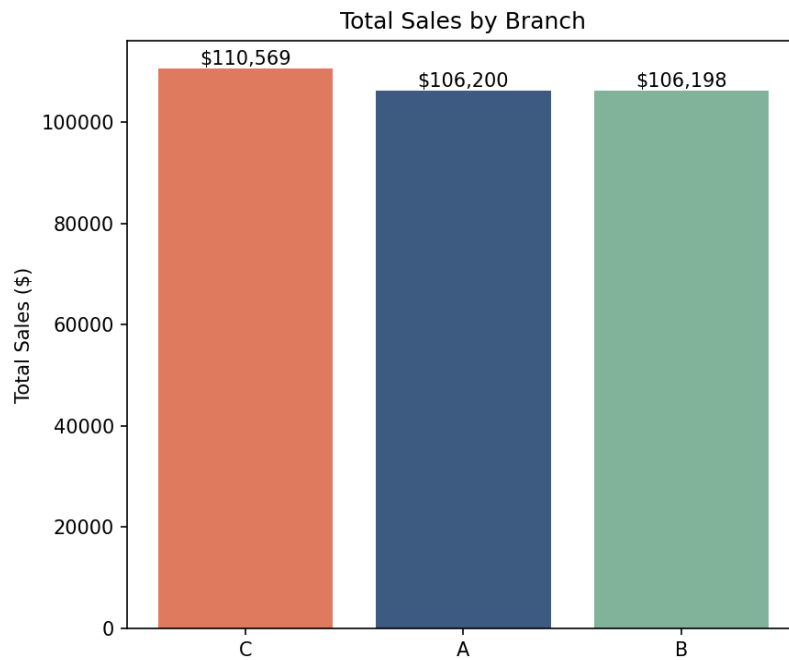


Figure 1: Total sales by branch.

Branch C brought in the most revenue at \$110,569, just ahead of Branch A (\$106,200) and Branch B (\$106,198), which are almost identical. Unlike some retail datasets where one location dramatically outperforms the others, these three branches are within about 4% of each other, suggesting a genuinely balanced customer base across locations rather than one dominant store.

3.2 Gender Splits Differ by Product Line



Figure 2: Purchases by product line, split by gender.

Most product lines are fairly balanced between male and female shoppers, but two stand out. Fashion accessories skews noticeably toward female customers, which isn't too surprising. What's more unexpected is Health and beauty, which actually skews toward male customers, the opposite of what

you might assume for that category. This is the kind of small anomaly that's easy to miss without actually plotting the data out.

3.3 No Single Payment Method Dominates

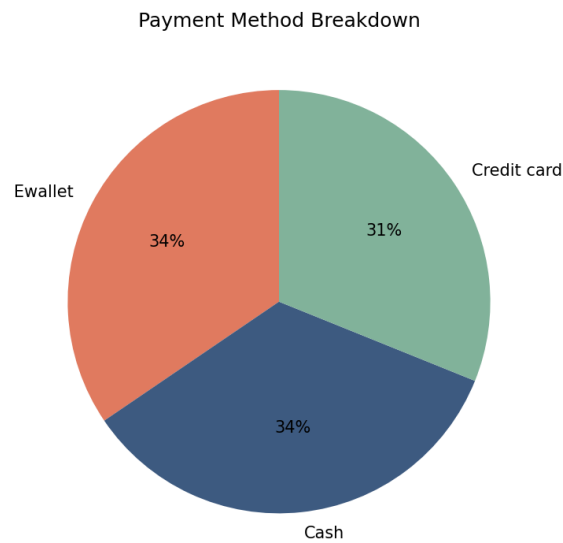


Figure 3: Payment method breakdown.

Customers split their payments almost evenly across the three available methods: Ewallet (34%), Cash (34%), and Credit card (31%). There's no clear favourite here, which is a bit different from what I'd expect, since usually one payment method tends to dominate. For this business, it likely means all three payment options need to stay well-supported, since customers don't lean heavily toward any one of them.

3.4 Saturday Is the Strongest Sales Day

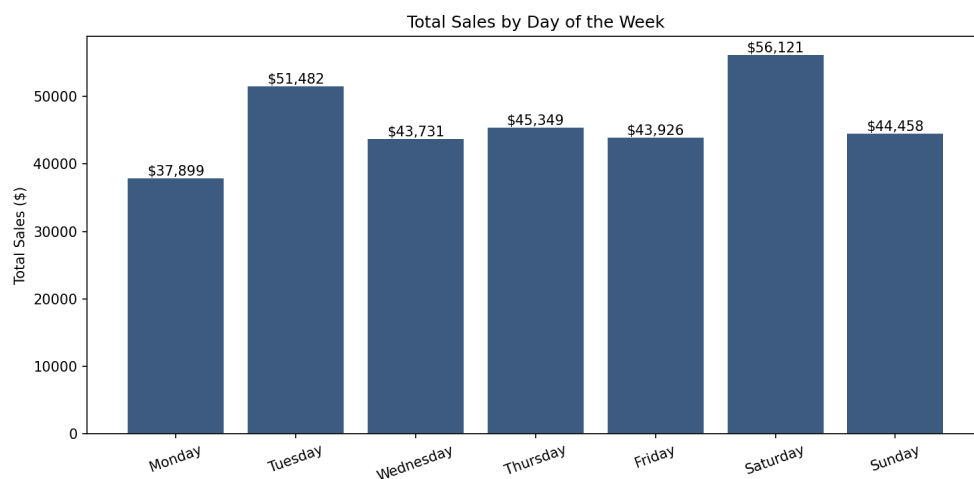


Figure 4: Total sales by day of the week.

Sales rise noticeably toward the weekend, peaking on Saturday at \$56,121, the highest of any day by a clear margin. Monday, on the other hand, is the weakest day at just \$37,899. This makes intuitive sense: people likely do more shopping when they're off work, and Saturday probably benefits from being the start of the weekend. For the business, this could mean adjusting staffing levels or running weekday promotions to smooth out the dip early in the week.

4. Conclusion

This analysis of the Supermarket Sales dataset turned up a few genuinely useful insights: the three branches perform at a similar level rather than one dominating; gender-based buying patterns differ by product line, with Health and beauty skewing unexpectedly male; customers don't strongly favour any one payment method; and sales build steadily toward a Saturday peak before dropping off again on Monday. Together, these patterns could support decisions around weekend staffing, payment infrastructure, and even how Health and beauty products are marketed.

5. Limitations

This dataset only covers three months (January to March 2019), so it's not possible to say whether these patterns hold up across a full year or reflect any seasonal effects. The data also comes from only three branches in one country, so the findings may not generalize to a larger or more diverse retail chain. A longer time frame and more locations would help confirm whether these trends are consistent or specific to this particular slice of data.